Theory of Mechanism and Machine

(Week-8)

Synthesis of Mechanism:

- *kinematic synthesis we mean the design or creation of a new mechanism to yield a desired set of motion characteristics.*
- *Kinematic synthesis of a mechanism requires the determination of lengths of various links that satisfy the requirements of motion of the mechanism.*
- *The usual requirements are related to specified positions of the input and output links. It is easy to design a planar mechanism, when the position of input and output links are known at fewer positions as compared to large number of positions.*
- *There are three general stages in the design of a new mechanism: type synthesis, number synthesis, and dimensional synthesis.*

- Type synthesis refers to the choice of the kind of mechanism to be used; it might be a linkage, a geared system, a cam system, or even belts and pulleys. This beginning stage of the total design process usually involves the consideration of design factors, such as manufacturing processes, materials, space, safety, and economics. The study of kinematics is usually only slightly involved in type synthesis.
- Number synthesis deals with finding a satisfactory number of links and number of joints to obtain the desired mobility . Number synthesis is the second stage in the design process and follows type synthesis.
- The third stage in design, determining the detailed dimensions of the individual links, is called dimensional synthesis.

Freudenstein equation :

The Freudenstein equation results from an analytical approach towards analysis and design of four-link mechanisms which, along with its variants, are present in a large number of machines used in daily life.

It relates the input crank angle, output rocker angle, and the lengths of the four links.

- Applications:

Position analysis of four-bar mechanisms.

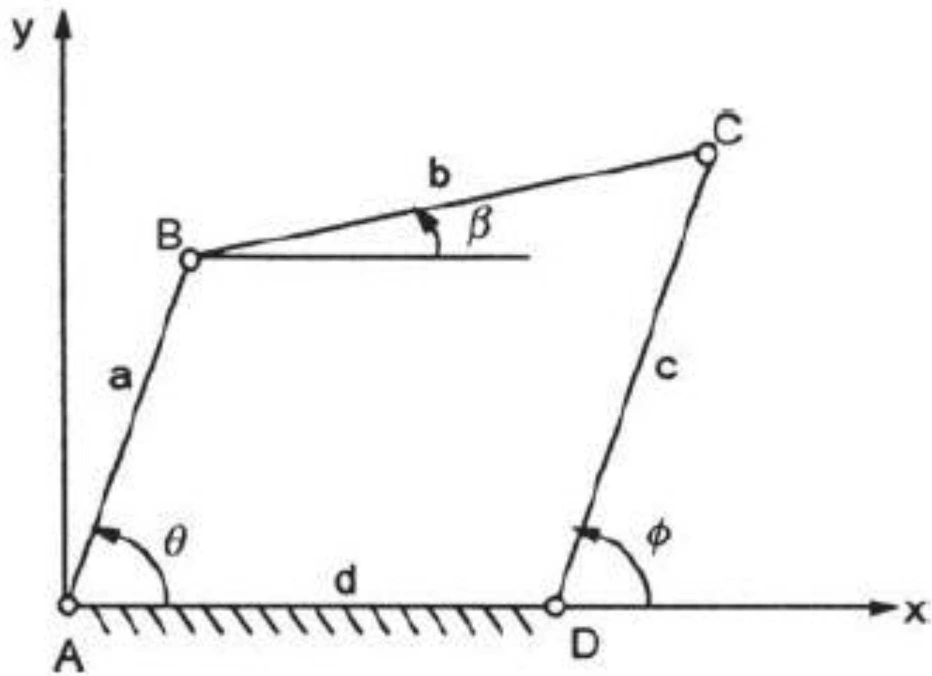
Function generation .

Path generation .

Motion generation .

Mechanism synthesis using precision points.

- *Consider a four-bar mechanism as shown in Fig. in equilibrium. The magnitudes of the links AB, BC, CD and DA are a, b, c and d respectively. θ, β and ϕ are the angles of AB, BC and DC respectively with the x-axis. AD is the fixed link. AB is the input link and DC the output link.*



The displacement along x-axis is:

$$a\cos\theta + b\cos\beta = d + c\cos\phi$$

$$\text{or} \qquad b\cos\beta = c\cos\phi - a\cos\theta + d$$

$$\text{or} \qquad \left(b\cos\beta\right)^2 = \left(c\cos\phi - a\cos\theta + d\right)^2$$

$$= c^2\cos^2\phi + a^2\cos^2\theta + d^2 - 2ac\cos\theta\cos\phi - 2ad\cos\theta + 2cd\cos\phi \quad (1)$$

The displacement along y-axis is:

$$a\sin\theta + b\sin\beta = c\sin\phi$$

$$b\sin\beta = c\sin\phi - a\sin\theta$$

$$\left(b\sin\beta\right)^2 = \left(c\sin\phi - a\sin\theta\right)^2$$

$$= c^2\sin^2\phi + a^2\sin^2\theta - 2ac\sin\theta\sin\phi \qquad (2)$$

Adding Eqs. (1) and (2), we get:

$$b^2 = c^2 + a^2 + d^2 - 2ac\,(\sin\theta\sin\phi + \cos\theta\cos\phi) - 2ad\cos\theta + 2cd\cos\phi$$

OR

$$2cd\cos\phi - 2ad\cos\theta + a^2 - b^2 + c^2 + d^2 = 2ac\,(\sin\theta\sin\phi + \cos\theta\cos\phi)$$

This is known as displacement equation.

Dividing throughout by 2*ac, we get;*

$$\frac{d}{a}\cos\phi - \frac{d}{c}\cos\theta + \frac{a^2 - b^2 + c^2 + d^2}{2ac} = \cos(\phi - \theta) = \cos(\theta - \phi)$$

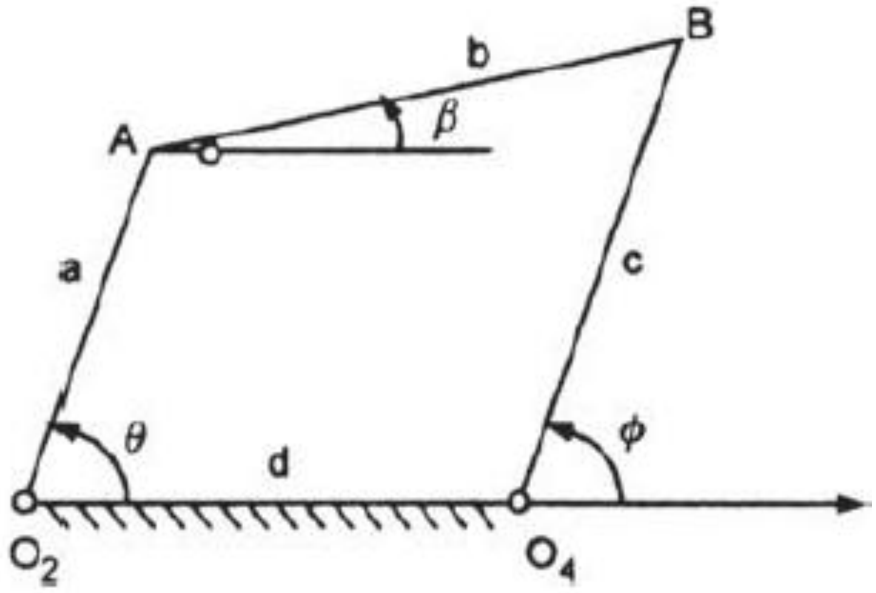
or $$k_1\cos\phi + k_2\cos\theta + k_3 = \cos(\theta - \phi)$$

where $$k_1 = \frac{d}{a},\ k_2 = \frac{-d}{c},\ \text{and}\ k_3 = \frac{a^2 - b^2 + c^2 + d^2}{2ac}$$

The above equation is known as Freudenstein equation.

Freudenstein's Equation for the Precision Points:

- Freudenstein's equation helps to determine the length of links of a four-bar mechanism. The displacement equation of a four-bar mechanism, shown in Fig. is given by-



$$2cd\cos\phi - 2ad\cos\theta + a^2 - b^2 + c^2 + d^2 = 2\,ac\,(\cos\theta\cos\phi + \sin\theta\sin\phi)$$

- Dividing throughout by 2*ac and rearranging, we get:*

$$\frac{d}{a}\cos\phi - \frac{d}{c}\cos\theta + \frac{a^2 - b^2 + c^2 + d^2}{2ac} = \cos(\theta - \phi)$$

- *Let;* $k_1 = \frac{d}{a}, k_2 = -\frac{d}{c}$ and $k_3 = \frac{a^2 - b^2 + c^2 + d^2}{2ac}$
- *Then;*

$$k_1 \cos\phi + k_2 \cos\theta + k_3 = \cos(\theta - \phi)$$

- *Let the input and output are related by some function, such as, y=f(x). For three specified positions, let*

$$\theta_1, \theta_2, \theta_3 = \text{three positions of input link}$$

$$\phi_1, \phi_2, \phi_3 = \text{three positions of output link}$$

The Freudenstein's equation can be written as:

$$k_1 \cos \phi_1 + k_2 \cos \theta_1 + k_3 = \cos(\theta_1 - \phi_1)$$
$$k_1 \cos \phi_2 + k_2 \cos \theta_2 + k_3 = \cos(\theta_2 - \phi_2)$$
$$k_1 \cos \phi_3 + k_2 \cos \theta_3 + k_3 = \cos(\theta_3 - \phi_3)$$

These equations can be written in the matrix form as:

$$\begin{bmatrix} \cos\phi_1 & \cos\theta_1 & 1 \\ \cos\phi_2 & \cos\theta_2 & 1 \\ \cos\phi_3 & \cos\theta_3 & 1 \end{bmatrix} \begin{Bmatrix} k_1 \\ k_2 \\ k_3 \end{Bmatrix} = \begin{Bmatrix} \cos(\theta_1 - \phi_1) \\ \cos(\theta_2 - \phi_2) \\ \cos(\theta_3 - \phi_3) \end{Bmatrix}$$

These equations can be solved by any numerical technique.

Using Cramer's rule, let

$$A=\begin{vmatrix}\cos\phi_1 & \cos\theta_1 & 1\\ \cos\phi_2 & \cos\theta_2 & 1\\ \cos\phi_3 & \cos\theta_3 & 1\end{vmatrix},\ A_1=\begin{vmatrix}\cos(\theta_1-\phi_1) & \cos\theta_1 & 1\\ \cos(\theta_2-\phi_2) & \cos\theta_2 & 1\\ \cos(\theta_3-\phi_3) & \cos\theta_3 & 1\end{vmatrix}$$

$$A_2=\begin{vmatrix}\cos\phi_1 & \cos(\theta_1-\phi_1) & 1\\ \cos\theta_2 & \cos(\theta_2-\phi_2) & 1\\ \cos\phi_3 & \cos(\theta_3-\phi_2) & 1\end{vmatrix},\ A_3=\begin{vmatrix}\cos\phi_1 & \cos\theta_1 & \cos(\theta_1-\phi_1)\\ \cos\phi_2 & \cos\theta_2 & \cos(\theta_2-\phi_2)\\ \cos\phi_3 & \cos\theta_3 & \cos(\theta_3-\phi_3)\end{vmatrix}$$

Then $k_1=\dfrac{A_1}{A},\ k_2=\dfrac{A_2}{A},\ k_3=\dfrac{A_3}{A}$

Knowing k1, k2 and k3, the values of a, b, c, and d can be calculated. Value of either 'a' or 'd' can be assumed to be unity to obtain the proportionate values of other parameters.

Q 1: Design a four-bar mechanism to coordinate three positions of the input and output links given by:

$$\theta_1 = 25, \phi_1 = 30°; \theta_2 = 35°, \phi_2 = 40°; \theta_3 = 50°, \phi_3 = 60°$$

Solution:

$$\cos\theta_1 = \cos 25° = 0.9063, \cos\theta_2 = \cos 35° = 0.8191, \cos\theta_3 = \cos 50° = 0.6428$$

$$\cos\phi_1 = \cos 30° = 0.8660, \cos\phi_2 = \cos 40° = 0.7660, \cos\phi_3 = \cos 60° = 0.5000$$

$$\cos(\theta_1 - \phi_1) = \cos(25° - 30°) = 0.9962$$

$$\cos(\theta_2 - \phi_2) = \cos(35° - 40°) = 0.9962$$

$$\cos(\theta_2 - \phi_2) = \cos(50° - 60°) = 0.9848$$

$$A = \begin{vmatrix} \cos\phi_1 & \cos\theta_1 & 1 \\ \cos\phi_2 & \cos\theta_2 & 1 \\ \cos\phi_3 & \cos\theta_3 & 1 \end{vmatrix}$$

$$A = \begin{vmatrix} 0.8660 & 0.9063 & 1 \\ 0.7660 & 0.8191 & 1 \\ 0.5000 & 0.6428 & 1 \end{vmatrix}$$

$$= 0.8660\,(0.8191 - 0.6428) - 0.9063\,(0.7660 - 0.5000) + 1(0.7666 \times 0.6428 - 0.8191 \times 0.5000)$$

$$= 5.5652 \times 10^{-3}$$

$$A_1 = \begin{vmatrix} \cos(\theta_1 - \phi_1) & \cos\theta_1 & 1 \\ \cos(\theta_2 - \phi_2) & \cos\theta_2 & 1 \\ \cos(\theta_3 - \phi_3) & \cos\theta_3 & 1 \end{vmatrix} \qquad A_1 = \begin{vmatrix} 0.9962 & 0.9063 & 1 \\ 0.9962 & 0.8191 & 1 \\ 0.9848 & 0.6428 & 1 \end{vmatrix}$$

$$= 0.9962\,(0.8191 - 0.6428) - 0.9063\,(0.9962 - 0.9848) + 1(0.9962 \times 0.6428 - 0.8191 \times 0.9848)$$

$$= -9.9408 \times 10^{-4}$$

$$A_2 = \begin{vmatrix} \cos\phi_1 & \cos\left(\theta_1 - \phi_1\right) & 1 \\ \cos\theta_2 & \cos\left(\theta_2 - \phi_2\right) & 1 \\ \cos\phi_3 & \cos\left(\theta_3 - \phi_2\right) & 1 \end{vmatrix} \qquad A_2 = \begin{vmatrix} 0.8660 & 0.9063 & 1 \\ 0.7660 & 0.9962 & 1 \\ 0.5000 & 0.9848 & 1 \end{vmatrix}$$

$$= 0.8660\,(0.9962 - 0.9848) - 0.9962\,(0.7660 - 0.5000) + 1(0.7660 \times 0.9848 - 0.9962) \times 0.5000)$$

$$= -1.14 \times 10^{2}$$

$$A_3 = \begin{vmatrix} \cos\phi_1 & \cos\theta_1 & \cos(\theta_1 - \phi_1) \\ \cos\phi_2 & \cos\theta_2 & \cos(\theta_2 - \phi_2) \\ \cos\phi_3 & \cos\theta_3 & \cos(\theta_3 - \phi_3) \end{vmatrix}$$

$$A_3 = \begin{vmatrix} 0.8660 & 0.9962 & 0.9962 \\ 0.7660 & 0.8191 & 0.9962 \\ 0.5000 & 0.6428 & 0.9848 \end{vmatrix}$$

$$= 0.8660\,(0.8191 \times 0.9848 - 0.9962 \times 0.6428) - 0.9063\,(0.7660 \times 0.9848 - 0.9962 \times 0.5000)$$

$$= 5.746 \times 10^{-3}$$

$$k_1 = \frac{A_1}{A} = \frac{-9.9408 \times 10^{-4}}{-5.5652 \times 10^{-3}} = 0.1786 = \frac{d}{a}$$

Let d=1 unit, then a=5.598 units

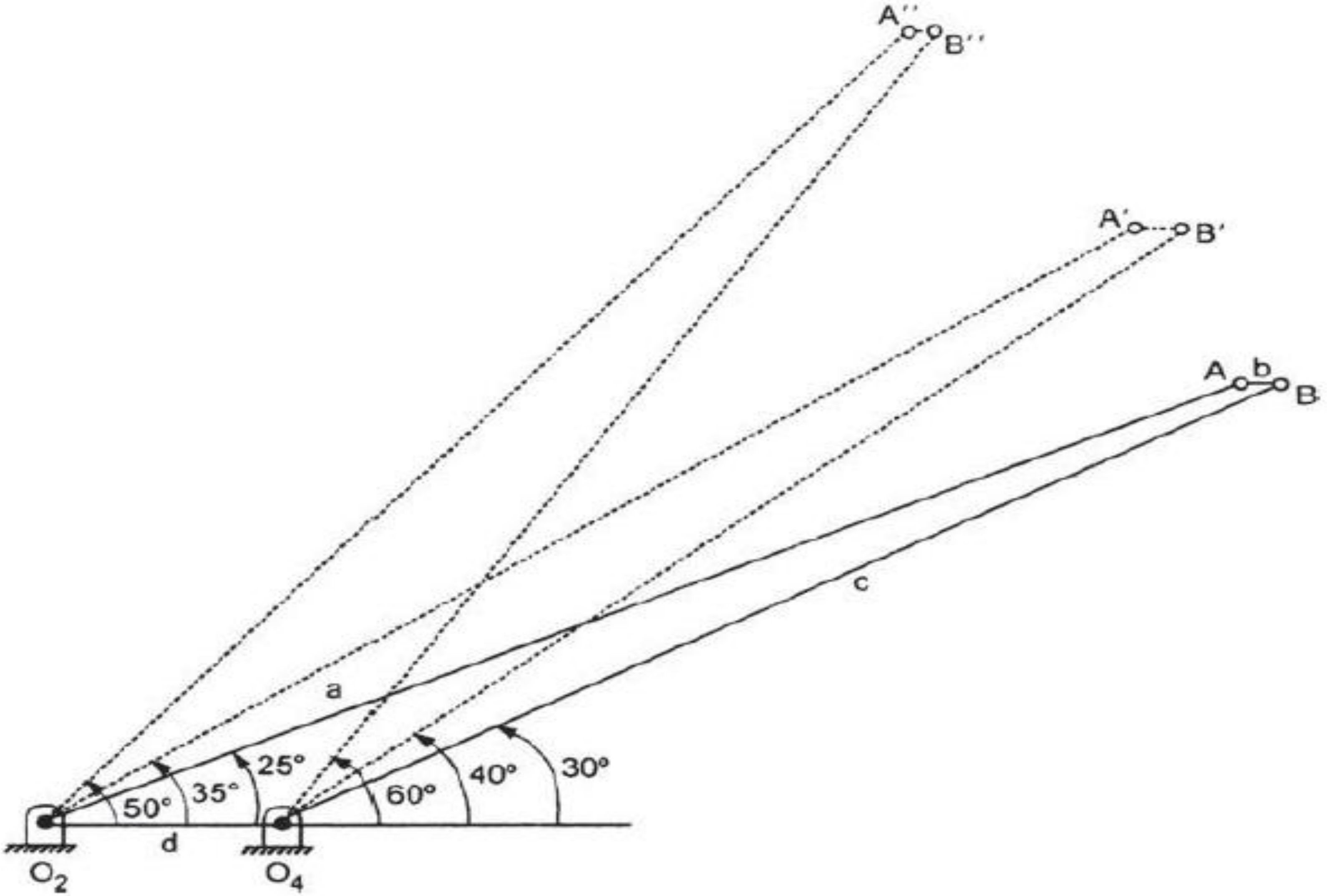
$$k_2 = \frac{A_2}{A} = \frac{1.14 \times 10^{-3}}{-5.5652 \times 10^{-3}} = -0.2048$$

$$= -\frac{d}{c}, \; c = 4.88 \text{ units}$$

$$k_3 = \frac{A_3}{A} = \frac{-5.764 \times 10^{-3}}{-5.5652 \times 10^{-3}} = 1.0272$$

$$= \frac{a^2 - b^2 + c^2 + d^2}{2ac}$$

$$= \frac{(5.598)^2 = b^2 + (4.88)^2}{2 \times 5.598 \times 4.88}, \qquad b = 0.176$$



Four-bar mechanism developed by three position precision points